

TSA Meeting Preparation — Detailed Study Note

Slide 12: Ch 8 — Microgrid Mode-Switching Protection

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Sources: SAMBP TR-38, TR-39, TR-40, TR-41, TR-42 / 2026

1. Slide Overview and Purpose

What This Slide Is About

This slide addresses one of the **most critical reliability gaps** in IBR microgrid protection: when a microgrid transitions from *grid-connected* to *islanded* mode, the fault current magnitude drops by **64–72%** and the current direction reference changes. Legacy relays designed for the grid-connected state will either **mal-operate** or **fail to detect** faults in the islanded state.

The slide presents the SAMBP solution: a **Bayesian mode detector** that continuously classifies the operating state, and an **adaptive directional element** that reconfigures itself in real time based on the detected mode.

Position in the thesis: Chapter 5 (Microgrid Protection), slide 12 of 22 in the TSA presentation. This is the second of two microgrid slides, following the GFL/GFM fault current characterisation (slide 11).

2. The Engineering Challenge

Why the mode transition is dangerous

When the Point-of-Common-Coupling (PCC) breaker opens (intentional or unintentional islanding), the following changes occur simultaneously:

Parameter	Grid-Connected	Islanded
Fault current source	Upstream grid infeed ($\gg$ IBR)	IBR only (limited to 1.1–2.0 pu)
Fault current magnitude	14–22 pu (bolted)	4–8 pu (IBR-limited)
Direction reference	Upstream grid voltage	GFM inverter terminal voltage
Frequency regulator	Stiff grid (50 Hz)	GFM virtual inertia ($H_{eq} = 0.5$ s)
Distance relay validity	Valid (remote infeed)	Invalid (no remote reference)
87L validity	Valid (comms path open)	Invalid (PCC breaker severs link)

The 22% Mal-Operation Problem

In the SAMBP study, without mode-adaptive protection, **22%** of fault events during the grid-to-island transition window result in relay mal-operation (either missed trip or false trip). This is the key “before” number that justifies the entire microgrid

protection chapter.

3. The Bayesian Mode Detector

Mathematical formulation

The operating mode is classified at each sample using a Bayesian maximum likelihood estimator:

Mode Detector Formula (slide equation)

$$\hat{m} = \arg \max_{m \in \{G, I\}} \Pr(z_t \mid \text{mode} = m, \hat{k}_{\text{ibr}}) \quad (1)$$

where:

- $\hat{m}$ — estimated mode: G = grid-connected, I = islanded
- z_t — measurement vector at time t (frequency deviation, ROCOF, voltage magnitude, reactive-power perturbation response)
- $\hat{k}_{\text{ibr}}$ — real-time IBR penetration estimate from the EKF digital twin (Chapter 6 / C3)
- $\Pr(\cdot)$ — conditional likelihood, updated at each 50 Hz sample

Why Bayesian, not a simple threshold?

A fixed ROCOF threshold (IEEE 1547-2018: ≥ 1 Hz/s) has a Non-Detection Zone (NDZ) of $\pm 2\%$ rated power mismatch. The Bayesian estimator fuses four signals:

1. **ROCOF** — primary passive signal, ≥ 1 Hz/s confirms islanding for $|\Delta P| \geq 0.02$ pu within 100 ms
2. **Voltage/frequency window** — secondary check ($V \notin [0.88, 1.10]$ pu or $f \notin [49.0, 50.5]$ Hz)
3. **Active ΔQ injection** — 0.05 pu reactive pulse; grid-connected bus shows $\Delta f < 0.01$ Hz; islanded GFM shows $\Delta f = 0.10$ Hz
4. $\hat{k}_{\text{ibr}}$ **from EKF** — contextual prior (high k_{ibr} means islanding is more likely given a given ROCOF value)

The active Q-injection reduces the NDZ area by **87%** (from $\pm 2\% \times \pm 5\%$ to $\pm 1.2\% \times \pm 1.5\%$ in ΔP – ΔQ space).

4. Adaptive Directional Element (67)

The slide lists two operating modes of the directional element:

Grid-connected mode (G)

- Polarising quantity: **pre-fault load flow voltage** from the upstream network — same as a conventional 67 relay
- Forward direction: toward the protected line (away from source)
- Reverse direction: toward the upstream grid
- Active elements: Z1/Z2 distance, 87L, 67Q/67N, Z2/POTT

- Pickup: $I_p = 1.50$ pu (Group 1 OC setting)

Islanded mode (I)

- Polarising quantity: **GFM inverter terminal voltage** (positive-sequence V_1 maintained by the GFM virtual impedance ≥ 0.5 pu for all faults within the 1.1–1.5 pu current limit)
- Why: The upstream grid reference disappears when the PCC breaker opens; the GFM virtual impedance provides a stable local voltage reference
- Active elements: 87B (bus differential, always on), OC Group 2, 67 directional
- Disabled: Z1/Z2 distance, 87L, Z2/POTT (all require remote reference)
- Pickup: $I_p = 0.30$ pu (Group 2, $5\times$ lower than grid-connected)
- **Key result:** OC Group 2 is $8\times$ more sensitive than 87L ($k_{ibr,min} = 0.0075$ pu vs 0.06 pu)

The 20 ms mode-switch window

Mode-switching is triggered by an **IEC 61850 GOOSE** signal from the TR-39 islanding relay, delivered within 4 ms of the islanding decision. The full sequence:

Event	Time	Protection state
PCC breaker opens (islanding)	$t = 0$ ms	Full grid-connected scheme
ROCOF reaches 1 Hz/s	$t = 100$ ms	ROCOF trip issued
GOOSE mode-switch signal	$t = 104$ ms	Within 4 ms of ROCOF trip
Setting group change complete	$t = 120$ ms	Islanded scheme active
Transition gap	20 ms	87B + 87L + OC-G1 still covering

Zero protection gap: during the 20 ms window, 87B (always active) and 87L (still enabled pre-switch) provide continuous fault coverage.

5. Results — What the Table in the Slide Means

Mode-Switch Results Table Explained			
Metric	Fixed	SAMBP-MG	Explanation
Mode detection acc.	—	98.7%	Bayesian detector correctly identifies G vs I in 98.7% of events (across all ΔP , ΔQ combinations in the study)
Transition mal. rate	22%	1.4%	Relay mal-operations (false trip or missed trip) during the grid $\leftrightarrow$ island transition; reduced by 15 $\times$
Islanded fault TPR	71%	97.3%	True positive rate for faults occurring in islanded mode; gap closed by OC Group 2 (lower pickup)
False islands	6/100	0/100	Events where the scheme incorrectly declared islanding during normal operation; eliminated by the multi-signal Bayesian fusion

The headline observation

The slide states: “Mode-adaptive protection eliminates the 22% mis-operation rate seen at grid $\leftrightarrow$ island transition boundary.”

This is the **single most important result** of this slide. The 22% figure represents a very high risk for conventional relays and is the quantitative justification for the entire adaptive microgrid protection framework.

6. Best Supporting Figure — TR-39 Figure 3

Recommended Figure for the Presentation	
Source: SAMBP TR-39/2026, Figure 3 (page 3)	
Title: Protection mode switching state machine	
The figure shows a state machine with five states:	
1. Grid-Connected Mode — full scheme: Z1/Z2 + 87L + 67Q active, OC at grid level	

2. **Islanding Detection** — ROCOF/V/f monitoring active
3. **Islanded Mode** — Z1/Z2 disabled, 87L offline, OC rescaled, GFM: V/f regulation
4. **Resyncing (Relay 25)** — $\Delta V \leq 0.05$, $\Delta f \leq 0.10$ Hz, $\Delta \theta \leq 10^\circ$
5. **Grid-Connected (restored)** or **Lockout** (resync failed)

Timing annotations: detection $t_{det} = 100$ ms, mode switch $t_{sw} = 120$ ms from islanding, Relay 25 window ≤ 5 s.

Why it is the best match: It directly visualises the mode-transition logic and timing that underlies every result in the slide table. It is self-contained and immediately comprehensible to a TSA committee audience.

Secondary figure (TR-40, Figure 3): The three-state adaptive OC setting group state machine (Group 1 GC → Group 2 Islanded → Group 3 Resync) is an excellent companion to explain the adaptive directional element bullet points.

7. Connections Across the Thesis

Chapter/Section	Connection to this slide
Ch 2 (IBR Foundations)	GFL/GFM fault current model justifies the 64–72% reduction figure
Ch 3 (Differential)	87L is <i>disabled</i> in islanded mode (this slide); 87B replaces it
Ch 4 (OC/Distance)	Distance relay Z1/Z2 also <i>disabled</i> in islanded mode; OC Group 2 takes primary role
Ch 6 (EKF Digital Twin)	$\hat{k}_{ibr}$ estimate feeds directly into the Bayesian mode detector (Eq. 1)
Ch 7 (WAPC)	IEC 61850 GOOSE is the triggering mechanism for the setting group change (4 ms delivery)

8. Equations Quick Reference

Key Equations for This Slide

ROCOF-based islanding detection (TR-39):

$$\text{ROCOF} = \frac{|\Delta P| \cdot f_0}{2H_{eq}} \geq 1 \text{ Hz/s} \Rightarrow |\Delta P| \geq 0.02 \text{ pu} \quad (2)$$

Active Q-injection frequency response (TR-39):

$$\Delta f = K_{f,Q} \cdot \Delta Q = 2.0 \times 0.05 = 0.10 \text{ Hz} \quad (\text{detectable vs. floor } \Delta f_{resol} = 0.01 \text{ Hz}) \quad (3)$$

Islanded OC minimum sensitivity (TR-40):

$$k_{\min,OC} = \frac{I_{p,2} \times Z_{\text{line}}}{2} = \frac{0.30 \times 0.050}{2} = 0.0075 \text{ pu} \quad (8\times \text{ more sensitive than 87L}) \quad (4)$$

Resync transient energy check (TR-39):

$$\Delta W = \frac{1}{2} S_{ibr} (1 - \cos \Delta \theta_{\max}) = 0.0076 \text{ pu}\cdot\text{s} \ll \Delta W_{\text{limit}} \approx 0.05 \text{ pu}\cdot\text{s} \quad (5)$$

9. TSA Meeting Speaker Script**What to Say at the TSA Meeting**

“This slide addresses what I call the **mode-transition blind spot** — the period when the microgrid is switching from grid-connected to islanded operation. During this window, legacy relays face a dilemma: they were set for high grid-fed fault currents, but now the fault current has dropped by 64 to 72 percent, and the directional reference has moved from the upstream grid to the GFM inverter terminal.

In our study, without adaptation, **22 percent** of events at the transition boundary resulted in relay mal-operation. That is an unacceptably high failure rate.

Our solution has two parts. First, a **Bayesian mode detector** that fuses four signals — ROCOF, voltage-frequency window, an active reactive power perturbation, and the IBR penetration estimate from the digital twin — to classify the operating mode at every sample. Second, an **adaptive directional element** that automatically switches from pre-fault load-flow polarisation to GFM terminal voltage polarisation when islanding is confirmed.

The result: the mal-operation rate falls from 22 percent to **1.4 percent**. Islanded fault detection improves from 71 to **97.3 percent**. And false island declarations — which would cause unnecessary outages — drop from 6 to **zero** in a hundred test events.

The switching is triggered by an IEC 61850 GOOSE signal, delivered within 4 milliseconds of the islanding decision, with a total transition time of 20 milliseconds. During that window, the bus differential element provides uninterrupted protection coverage — so there is **zero protection gap**.”

10. Anticipated TSA Committee Questions and Answers**1. Q: How is the Bayesian mode detector different from a simple ROCOF relay?**

A: A simple ROCOF relay has a fixed NDZ of $\pm 2\%$ rated power. Our Bayesian detector fuses ROCOF with active Q-injection and the EKF’s $\hat{k}_{ibr}$ estimate, reducing the NDZ area by 87% and eliminating all false declarations.

2. Q: What happens if the GOOSE signal is delayed or lost?

A: The 87B bus differential is always active in both modes and requires no GOOSE signal. It covers busbar faults during the transition. For the 20 ms gap, the pre-switch settings (OC Group 1 + 87L) remain active, providing backup coverage.

3. Q: The 1.4% residual mal-operation rate — what causes it?

A: Both residual misses are *structural* — they occur when $k_{ibr} < 0.0075$ pu, below the absolute detection floor of the islanded OC element. This corresponds to an inverter operating below 0.75% of rated current, which is outside the scope of any

protection scheme. The grid-code recommendation is $k_{ibr} \geq 0.10$ pu per unit.

4. **Q: Is hardware-in-the-loop validation planned?**

A: Yes. The identified next step is HIL testing on a commercial IED (SEL-411L) with a real-time PSCAD simulation via the DRTS interface.

5. **Q: How does this interact with the WAPC in Chapter 7?**

A: The WAPC receives the islanding status via IEC 61850 MMS and updates the wide-area settings accordingly. The GOOSE mode-switch signal originates at the local islanding relay (TR-39) and is also broadcast to the WAPC for N-1 contingency re-assessment.

Study note prepared from SAMBP Technical Reports TR-38 through TR-42 (2026).

Best supporting figure: TR-39/2026, Figure 3 — Protection Mode Switching State Machine.

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